

# Bibas Pokhrel

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## PROFESSIONAL EXPERIENCE

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### Water Resources Intern | *Hanson Professional Services, Springfield, Illinois* | May 2026 – August 2026

- Developed a Python–ArcGIS Pro pipeline for IDOT's BridgeWatch early-warning system, assessing flood-monitoring coverage for 650+ Illinois bridges using USGS streamgages and NEXRAD data and translating results into monitoring and field-calibration recommendations.
- Performed watershed delineation, land-cover classification, curve-number generation, and basin-parameter extraction in ArcGIS Pro; applied HydroCAD, Excel, and OpenRoads for highway drainage design, culvert sizing, and overtopping assessments.
- Built an SRH-2D hydraulic model with HY-8 culvert structures for a critical interstate crossing, including terrain processing, material classification, breaklines, mesh generation, boundary conditions, and 2D flow analysis.
- Built a 1D HEC-RAS model for a dam-removal project to evaluate hydraulic conditions and estimate accumulated sediment volume for design and planning.

### GIS Expert | *Tundi Constructions, Kathmandu, Nepal* | May 2024 – July 2025

- Managed GIS data for the Melamchi Water Supply Project, maintaining spatial integrity and asset records for 125+ miles of water-distribution pipelines.
- Produced engineering maps and recommendations, coordinated DGPS asset-verification surveys, and trained field teams in geospatial data collection and quality control.

## RESEARCH EXPERIENCE

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### Graduate Research Assistant | *Southern Illinois University Carbondale, Carbondale, Illinois* | Aug 2025 – Dec 2026

- Developed a 92-mi<sup>2</sup> HEC-RAS 2D rain-on-grid model using LiDAR, NLCD, SSURGO, and MRMS data to evaluate flood inundation and detention alternatives for a citywide stormwater master plan.
- Evaluated 18 Bulletin 75 design-storm scenarios to identify critical storm durations, assess level of service, locate flood-prone corridors and infrastructure at risk, and whether proposed BMP and detention-pond alternatives remain effective under more extreme storms.
- Developed a citywide PCSWMM model and Python-based workflow to evaluate five BMP practices, quantify peak-flow and runoff-volume reductions, and rank mitigation alternatives for capital planning.
- Evaluating Piles Fork Creek flooding and rehabilitation alternatives near SIU and University Hall, assessing creek-related flood risk and potential mitigation measures for vulnerable campus areas.

## SKILLS

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**Hydrologic & Hydraulic:** HEC-RAS 1D/2D, HEC-HMS, PCSWMM, SRH-2D, HY-8, HydroCAD

**GIS & Design:** ArcGIS Pro, ModelBuilder, ArcPy, Bentley OpenRoads, AutoCAD

**Programming & Data:** Python, GeoPandas, NumPy, Advanced Excel

**Engineering:** Watershed delineation, stormwater modeling, hydrologic/hydraulic modeling, roadway drainage, culvert sizing, floodplain mapping, flood inundation analysis, LID/BMP evaluation

## ACTIVITIES & LEADERSHIP

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### Communications Lead | *2026 ASCE Mid-America Student Symposium (SIUC & SIUE)* | August 2025 – April 2026

- Led communications and logistics for a three-day ASCE regional symposium serving 500+ students from 17 universities; recruited and coordinated 35+ PE judges while managing schedules, training, venues, and volunteer assignments.

## EDUCATION

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**Southern Illinois University Carbondale**

*M.S. Civil Engineering — Water Resources & Hydrology*

Carbondale, Illinois

*Aug 2025 – Dec 2026 (Expected)*

**Tribhuvan University**

*B.E. Civil Engineering*

Kathmandu, Nepal

*Aug 2023 (Graduated)*